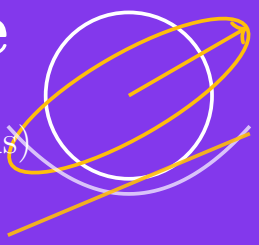
Linear Algebra, But Make It Make Sense

Lecture 6 – Stretch, shrink, converge, differentiate
(operator norms, low-rank SVD, sequences, $O/o/\Theta$, calculus)



Sasha Sutton – personal study notes, Fall 2026

My notes + proofs + fully solved exercises, remixed into one doc.

How to read this doc (the box legend)

DEFINITION	The lore. Learn these word for word, examiners love them.
THEOREM	The canon. Things that are always true.
RECEIPTS	Proofs. Every claim needs receipts.
EXAMPLE	Worked examples with actual numbers.
ML SIDE QUEST	Why a data scientist should care.
EXAM CHEAT CODE	"If they ask you *this*, you do *that*". Step-by-step recipes.
COMMON L	Classic mistakes. Don't be that person.
TL;DR	The summary for when you're cramming at 2am.

Everything here is real maths, just explained like a friend would. The vibes are optional, the rigour is not.

Contents

1 The big picture (read this first) 2

2 Operator norms: the maximal stretch 2

2.1 Definition . . . 2

2.2 Why only the unit sphere matters . . . 3

2.3 The formulas for $\|A\|_1$ and $\|A\|_\infty$ . . . 4

2.4 Lipschitz constants and submultiplicativity . . . 4

3 Spectral norm and low-rank approximation via SVD 5

3.1 The spectral norm $= \sigma_1$ . . . 5

3.2 SVD as a sum of rank-one pieces . . . 6

3.3 Least squares with the pseudo-inverse . . . 7

4 Sequences: where is this going? 8

4.1 Definitions . . . 8

4.2 Proving convergence: build $N(\varepsilon)$ . . . 9

5 Asymptotic orders: O, o, Θ 10

6 Limits and continuity 12

7 Derivative and first-order Taylor 13

8 Gradient and directional derivatives 15

9 Jacobian and the differential 17

10 The chain rule (a.k.a. backprop) 18

11 The exercises, fully solved 20

11.1 Sequences and asymptotics (Exercises 1, 2, 8, 9, 10) . . . 20

11.2 Operator norms and SVD (Exercises 3–7) . . . 23

11.3 Limits, continuity, derivatives (Exercises 11–14) . . . 25

11.4 Bonus: extra practice questions . . . 28

12 Final boss: the cheat sheet 28

12.1 "If they ask you this, do that" . . . 29

12.2 Formula sheet . . . 30

12.3 Hall of Ls . . . 30

12.4 The whole lecture in 10 lines . . . 31

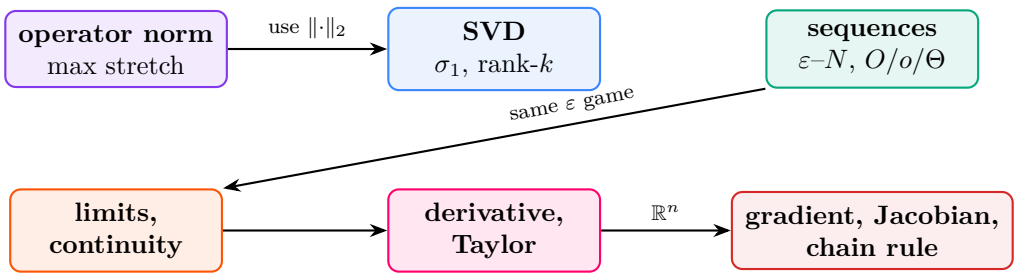
1 The big picture (read this first)

Previously on Linear Algebra

- **Lectures 1–4:** vector spaces, matrices, systems, determinants, eigenvalues.
- **Lecture 5:** inner products, norms, projections, spectral theorem, SVD.
- **Today:** two halves. First we finish the matrix story: *how much can a matrix stretch things*, and *how to keep only the important part of a matrix*. Then we switch to the calculus toolkit every ML model runs on: sequences, limits, derivatives, gradients, Jacobians, the chain rule.

TL;DR // The whole lecture as a story

1. The **operator norm** $\|A\|_{\text{op}}$ = the biggest factor by which A can stretch a vector. It's a Lipschitz constant: it bounds how much errors get amplified.
2. With the Euclidean norm, that's the **spectral norm** $\|A\|_2 = \sigma_1$, the top singular value.
3. **Eckart–Young:** chop the SVD after k terms and you get the *best* rank-k approximation. The error is the first singular value you threw away.
4. A **sequence converges** if its tail eventually lives in any tiny ball you choose (ε–N).
5. O, o, Θ compare how fast things grow or shrink: $\log n \ll n^\alpha \ll \beta^n$.
6. **Limits and continuity**: same ε game, but with functions (ε–η).
7. The **derivative** is the slope of the best straight-line zoom of a function: $f(x+h) \approx f(x) + f'(x)h$.
8. In many variables: the **gradient** (one output) and the **Jacobian** (many outputs) are the best linear zooms. The **chain rule** multiplies Jacobians, and that's literally backpropagation.



Top row: matrices and sequences. Bottom row: calculus. They meet at the end: a Jacobian is a matrix, so its operator norm bounds how sensitive a model is.

2 Operator norms: the maximal stretch

2.1 Definition

Lecture 5 measured the size of a *vector*. Now: how "big" is a *matrix*? A matrix is a machine that eats vectors, so a natural answer is: **how much can it stretch a vector, at most?**

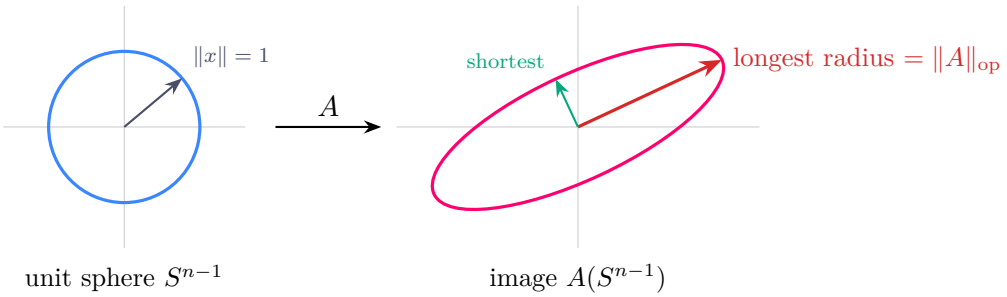
DEFINITION // Operator norm (a.k.a. induced or subordinate norm)

Fix a vector norm $\|\cdot\|$ on $\mathbb{R}^n$ (and one on $\mathbb{R}^m$). For $A \in \mathbb{R}^{m\times n}$:

$$\|A\|_{\text{op}} = \sup_{x\neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{\|x\|=1} \|Ax\|.$$

It is **subordinate** to $\|\cdot\|$, meaning: $\|Ax\| \le \|A\|_{\text{op}}\,\|x\|$ for every x.

Read it as: "feed A every vector of length 1; the longest thing that comes out has length $\|A\|_{\text{op}}$".



With the Euclidean norm, A turns the unit circle into an ellipse. The operator norm is the longest semi-axis.

THEOREM / PROPOSITION // It really is a norm (on matrices)

For all $A, B \in \mathbb{R}^{m\times n}$ and $\alpha \in \mathbb{R}$:

(i) $\|A\|_{\text{op}} \ge 0$, and $\|A\|_{\text{op}} = 0 \iff A = 0$;

(ii) $\|\alpha A\|_{\text{op}} = |\alpha|\,\|A\|_{\text{op}}$;

(iii) $\|A + B\|_{\text{op}} \le \|A\|_{\text{op}} + \|B\|_{\text{op}}$.

RECEIPTS (proof) //

(i) If $\|A\|_{\text{op}} = 0$, then $\|Ax\| = 0$ for every x, so $Ax = 0$ for all x, so $A = 0$ (look at Ae_j = column j). (ii) $\|\alpha Ax\| = |\alpha|\|Ax\|$, take the sup. (iii) For $\|x\| = 1$: $\|(A + B)x\| \le \|Ax\| + \|Bx\| \le \|A\|_{\text{op}} + \|B\|_{\text{op}}$. Take the sup over x. ■ *(and that's on periodt)*

2.2 Why only the unit sphere matters

THEOREM / PROPOSITION // Unit-sphere reduction

For $x \neq 0$, write $x = \|x\|\,u$ with $\|u\| = 1$. Then $\dfrac{\|Ax\|}{\|x\|} = \dfrac{\|x\|\,\|Au\|}{\|x\|} = \|Au\|$. So maximising the ratio over all $x \neq 0$ is the same as maximising $\|Au\|$ over unit vectors u. (And also over the unit *ball* $\|x\| \le 1$: see Exercise 4.)

EXAMPLE // Easy operator norms (any p-norm unless said)

1. I_n: does nothing, so $\|I_n\|_{\text{op}} = 1$.
2. αI_n: stretches everything by $|\alpha|$, so $\|\alpha I_n\|_{\text{op}} = |\alpha|$.
3. $D = \operatorname{diag}(d_1, \dots, d_n)$: stretches axis i by $|d_i|$, so $\|D\|_{\text{op}} = \max_i |d_i|$. E.g. $\|\operatorname{diag}(2, -5, 1)\|_{\text{op}} = 5$.

4. Orthogonal Q (Euclidean norm): rotations/reflections keep lengths, so $\|Q\|_2 = 1$.

2.3 The formulas for $\|A\|_1$ and $\|A\|_\infty$

THEOREM / PROPOSITION // Column sums and row sums

$$\|A\|_1 = \max_j \sum_i |a_{ij}| \quad \text{(max absolute \textbf{column} sum)},$$

$$\|A\|_\infty = \max_i \sum_j |a_{ij}| \quad \text{(max absolute \textbf{row} sum)}.$$

RECEIPTS (proof) // $\|A\|_1$ = max column sum

Let $c_j = \sum_i |a_{ij}|$ (column sums) and $c^\star = \max_j c_j$.

1. **Upper bound.** $\|Ax\|_1 = \sum_i |\sum_j a_{ij}x_j| \le \sum_i \sum_j |a_{ij}||x_j| = \sum_j |x_j|\, c_j$, and $\sum_j |x_j|\, c_j \le c^\star \sum_j |x_j| = c^\star \|x\|_1$.
2. **Attained.** Pick the best column $j^\star$ and $x = e_{j^\star}$ ($\|x\|_1 = 1$). Then Ax = column $j^\star$, of 1-norm $c^\star$.

So $\|A\|_1 = c^\star$. **For $\|A\|_\infty$:** $|(Ax)_i| \le \sum_j |a_{ij}||x_j| \le (\sum_j |a_{ij}|)\|x\|_\infty$, and equality is reached by the sign vector $x_j = \operatorname{sign}(a_{i^\star j})$ of the best row $i^\star$. ■ *(and that's on periodt)*

*Memory hook: 1 is a vertical stick, so **columns**; ∞ is lying down, so **rows**.*

EXAMPLE // Worked example: $A = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$

1. Column sums: $|2| + |1| = 3$, $|-1| + |3| = 4$. So $\|A\|_1 = 4$, attained at $x = e_2 = (0, 1)$: $Ax = (-1, 3)$, $\|Ax\|_1 = 4$ ✓.
2. Row sums: $|2| + |-1| = 3$, $|1| + |3| = 4$. So $\|A\|_\infty = 4$, attained at $x = (\operatorname{sign} 1, \operatorname{sign} 3) = (1, 1)$: $Ax = (1, 4)$, $\|Ax\|_\infty = 4$ ✓.
3. Bonus, the 2-norm: $A^\top A = \begin{pmatrix} 5 & 1 \\ 1 & 10 \end{pmatrix}$, $\lambda_{\max} = \frac{15+\sqrt{29}}{2} \approx 10.19$, so $\|A\|_2 \approx 3.19$. Same matrix, different geometry, different number. (Always $\|A\|_2 \le \sqrt{\|A\|_1 \|A\|_\infty}$: here $3.19 \le 4$ ✓.)

2.4 Lipschitz constants and submultiplicativity

THEOREM / PROPOSITION // Operator norm = Lipschitz constant

For all x, y: $\|Ax - Ay\| = \|A(x - y)\| \le \|A\|_{\text{op}} \|x - y\|$. So $\|A\|_{\text{op}}$ is the **smallest** constant L with $\|Ax - Ay\| \le L\|x - y\|$: a perturbation δx of the input is amplified by at most $\|A\|_{\text{op}}$.

THEOREM / PROPOSITION // Submultiplicativity

$\|AB\|_{\text{op}} \le \|A\|_{\text{op}} \|B\|_{\text{op}}$ (whenever AB makes sense).

RECEIPTS (proof) // stretch, then stretch again

For $x \neq 0$ with $Bx \neq 0$: $\dfrac{\|ABx\|}{\|x\|} = \dfrac{\|A(Bx)\|}{\|Bx\|} \cdot \dfrac{\|Bx\|}{\|x\|} \leq \|A\|_{\text{op}}\|B\|_{\text{op}}$. (If $Bx = 0$, the left side is 0.) Take the sup. ■ *(and that's on periodt)*

EXAMPLE // Submultiplicativity in numbers

$A = \text{diag}(2,1)$, $B = \text{diag}(1,3)$, Euclidean norm: $\|A\|_2 = 2$, $\|B\|_2 = 3$, product bound 6. But $AB = \text{diag}(2,3)$, so $\|AB\|_2 = 3 < 6$. The bound is often loose: A and B stretch *different* directions. It's tight when they stretch the same one (e.g. $A = B = \text{diag}(2,1)$: $\|A^2\|_2 = 4 = 2 \cdot 2$).

ML SIDE QUEST // Why $\|A\|_{\text{op}}$ shows up in training

- **Step sizes:** for an update like $x \mapsto x - \eta Ax$, picking $\eta < 1/\|A\|_{\text{op}}$ keeps each step from overshooting (Exercise 14 does the 1D version).
- **Deep nets:** a network is a composition of layers, so its Lipschitz constant is at most the product of the layers' norms. That's why spectral normalisation (Lecture 5) keeps each $\|W_\ell\|_2 \leq 1$.
- **Preprocessing:** if you scale features by a matrix A, $\|Ax\| \leq \|A\|_{\text{op}}\|x\|$ certifies the magnitudes stay in a known range.
- **Choice of geometry:** the operator norm depends on the vector norm you choose. You can pick a norm that matches what you care about (e.g. a perceptual norm for audio).

3 Spectral norm and low-rank approximation via SVD

3.1 The spectral norm $= \sigma_1$

DEFINITION // Spectral norm

For $A \in \mathbb{R}^{m \times n}$: $\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^\top A)}$.

THEOREM / PROPOSITION // The Euclidean operator norm is the spectral norm

$\sup_{\|x\|_2=1} \|Ax\|_2 = \sigma_1(A)$, attained at the first right singular vector v_1 (and $Av_1 = \sigma_1 u_1$).

RECEIPTS (proof) // rotate the problem away

1. SVD: $A = U\Sigma V^\top$ with U, V orthogonal (Lecture 5).
2. Substitute $x = Vy$. Since V is orthogonal, $\|x\|_2 = \|y\|_2$, so $\|x\|_2 = 1 \iff \|y\|_2 = 1$.
3. $\|Ax\|_2 = \|U\Sigma V^\top Vy\|_2 = \|U\Sigma y\|_2 = \|\Sigma y\|_2$ (U orthogonal keeps lengths).
4. $\|\Sigma y\|_2^2 = \sum_i \sigma_i^2 y_i^2 \leq \sigma_1^2 \sum_i y_i^2 = \sigma_1^2$.
5. Equality at $y = e_1$, i.e. $x = Ve_1 = v_1$. ■ *(and that's on periodt)*

THEOREM / PROPOSITION // Properties of $\|\cdot\|_2$

1. **Orthogonal invariance:** $\|QAR\|_2 = \|A\|_2$ for orthogonal Q, R (same singular values).
2. **Symmetric (or normal) A:** $\|A\|_2 = \max_i |\lambda_i|$ (Exercise 5).
3. **Sandwich with Frobenius:** $\|A\|_2 \le \|A\|_F \le \sqrt{\operatorname{rank} A}\,\|A\|_2$, because $\|A\|_F^2 = \sum_i \sigma_i^2$ (between σ_1^2 and $r\sigma_1^2$).

EXAMPLE // Worked example: $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

1. $A^\top A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, eigenvalues 5 ± 4: 9 and 1.
2. $\sigma_1 = 3$, $\sigma_2 = 1$, so $\|A\|_2 = 3$.
3. Top direction: $v_1 = \frac{1}{\sqrt{2}}(1,1)$. Check: $Av_1 = \frac{1}{\sqrt{2}}(3,3) = 3v_1$, length 3 ✓.
4. A is symmetric with eigenvalues $3, 1$, so $\max|\lambda_i| = 3$ too ✓.
5. $\|A\|_F = \sqrt{4+1+1+4} = \sqrt{10} \approx 3.16$, which sits between 3 and $\sqrt{2}\cdot 3$ ✓.

3.2 SVD as a sum of rank-one pieces

THEOREM / PROPOSITION // SVD recap

$A = U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i\, u_i v_i^\top$, $\sigma_1 \ge \cdots \ge \sigma_r > 0$, $r = \operatorname{rank} A$. Each $u_i v_i^\top$ is a rank-one matrix: "read direction v_i in the input, write direction u_i in the output, with volume σ_i". Also $\|A\|_F^2 = \sum_i \sigma_i^2$ (the "energy").

DEFINITION // Truncated SVD

For $k < r$: $\quad A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^\top = U_k \Sigma_k V_k^\top$ (keep the top k singular triplets).

THEOREM / PROPOSITION // Eckart–Young(–Mirsky)

A_k is the best approximation of A by a matrix of rank $\le k$, in both norms:

$$\min_{\operatorname{rank} B \le k} \|A - B\|_2 = \|A - A_k\|_2 = \sigma_{k+1}, \qquad \min_{\operatorname{rank} B \le k} \|A - B\|_F = \|A - A_k\|_F = \Big(\sum_{i>k} \sigma_i^2\Big)^{1/2}.$$

(Full proof: Exercise 6.) *The spectral error is the first singular value you threw away; the Frobenius error is all of them together.*

EXAM CHEAT CODE // "Give the best rank-k approximation and its error"

1. Get the SVD (eigen-decompose $A^\top A$ as in Lecture 5).
2. $A_k = \sum_{i \le k} \sigma_i u_i v_i^\top$.
3. Spectral error $= \sigma_{k+1}$. Frobenius error $= \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2}$.

4. Energy kept $= \dfrac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_r^2}$.

EXAMPLE // Rank-1 approximation of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

1. $\sigma_1 = 3$, $u_1 = v_1 = \frac{1}{\sqrt{2}}(1,1)$.
2. $A_1 = 3\, v_1 v_1^\top = 3 \cdot \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.
3. $A - A_1 = \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} = \sigma_2 u_2 u_2^\top$ with $u_2 = \frac{1}{\sqrt{2}}(1,-1)$.
4. Errors: $\|A - A_1\|_2 = \sigma_2 = 1$ and $\|A - A_1\|_F = \sqrt{4 \cdot 0.25} = 1$. (Careful: it's $\sqrt{0.25 + 0.25 + 0.25 + 0.25} = 1$, not $\sqrt{2}$; Eckart–Young says it must equal $\sigma_2 = 1$.)
5. Energy kept: $\frac{9}{9+1} = 90\%$.

EXAMPLE // Choosing k: singular values $5, 3, 1$

Say A has $\sigma = (5, 3, 1)$, total energy $25 + 9 + 1 = 35$.

k	$\|A - A_k\|_2$	$\|A - A_k\|_F$	energy kept
1	3	$\sqrt{9+1} = \sqrt{10} \approx 3.16$	$25/35 \approx 71\%$
2	1	1	$34/35 \approx 97\%$
3	0	0	100%

Here $k = 2$ keeps 97% for two thirds of the cost: the classic "elbow" choice.

3.3 Least squares with the pseudo-inverse

THEOREM / PROPOSITION // Least squares via SVD

Problem: $x^\star = \arg\min_x \|Ax - b\|_2$. With $A = U\Sigma V^\top$ and the **pseudo-inverse** $A^+ = V\Sigma^+ U^\top$ (Σ^+: invert the non-zero σ_i, transpose the shape):

1. $x^\star = A^+ b$ is a least-squares solution;
2. $\hat{b} = Ax^\star = U_r U_r^\top b$ is the orthogonal projection of b onto range(A) (Lecture 5); the residual is in range$(A)^\perp$;
3. among all least-squares solutions, $x^\star$ has the **smallest norm**.

EXAMPLE // A rank-deficient system

$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $b = (1, 3)$. No exact solution ($1 \neq 3$), and infinitely many least-squares ones.

1. SVD: $\sigma_1 = 2$, $u_1 = v_1 = \frac{1}{\sqrt{2}}(1,1)$, $\sigma_2 = 0$.
2. $A^+ = \frac{1}{\sigma_1} v_1 u_1^\top = \frac{1}{2} \cdot \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

3. $x^\star = A^+ b = \frac{1}{4}(4,4) = (1,1)$.
4. $\hat{b} = Ax^\star = (2,2)$: the projection of $(1,3)$ onto the line span$(1,1)$ (the average). Residual $(-1,1) \perp (1,1)$ ✓.
5. All least-squares solutions: $x_1 + x_2 = 2$. The closest to 0 on that line is $(1,1)$ ✓ (minimal norm).

DEFINITION // Condition number

For full-rank A: $\kappa_2(A) = \sigma_1/\sigma_r$. It measures how much relative errors in b can be amplified in x.

EXAMPLE // Ill-conditioning in one line

$A = \mathrm{diag}(1,\ 0.001)$: $\kappa = 1000$. Solving $Ax = b$: a tiny error 0.001 in b_2 becomes an error $0.001/0.001 = 1$ in x_2. **Truncated SVD** ($x_k = V_k \Sigma_k^{-1} U_k^\top b$) drops the tiny σ_i to avoid this, at the cost of a bit of bias.

ML SIDE QUEST // Low rank is everywhere

- **PCA:** project centred data on the top singular vectors. **Centre first**, otherwise v_1 just points at the mean.
- **Recommender systems / word embeddings:** approximate user–item or word–context matrices by low rank to find hidden factors.
- **Compression:** replace a dense $m \times n$ layer by $U_k \Sigma_k V_k^\top$: $k(m+n+1)$ numbers instead of mn. For $m = n = 1000$, $k = 50$: $\approx$ 100,000 instead of 1,000,000.
- **LoRA:** fine-tune a big language model by learning a low-rank *update* $\Delta W = BA$ with $B \in \mathbb{R}^{m \times k}$, $A \in \mathbb{R}^{k \times n}$, k small. Much less memory than updating all of W.

4 Sequences: where is this going?

TL;DR // Why a data scientist cares

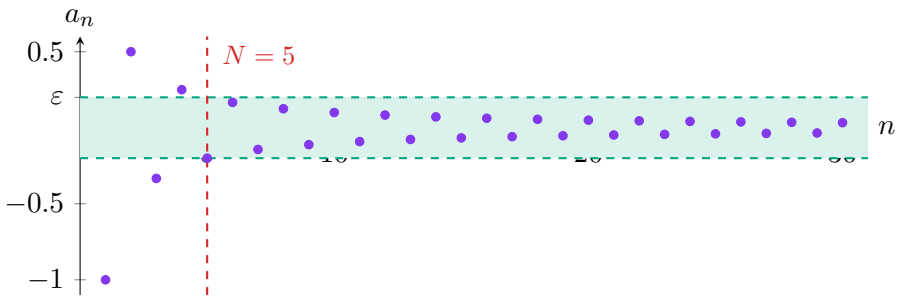
Training is a sequence: $w_0, w_1, w_2, \ldots$ of weights, and $f(w_0), f(w_1), \ldots$ of losses. "Does it converge, and how fast?" is the question behind every training curve.

4.1 Definitions

DEFINITION // Convergence, divergence, Cauchy (in $(\mathbb{R}^d, \|\cdot\|)$)

- $(x_n) \to x^\star$ iff for every $\varepsilon > 0$ there is N such that $n \geq N \Rightarrow \|x_n - x^\star\| < \varepsilon$.
- (Real sequences) $x_n \to +\infty$ iff for every $A > 0$ there is N such that $n \geq N \Rightarrow x_n \geq A$. (Same for $-\infty$ with $x_n \leq -A$.)
- (x_n) is **Cauchy** iff for every $\varepsilon > 0$ there is N such that $m, n \geq N \Rightarrow \|x_n - x_m\| < \varepsilon$.

In words: you pick a tolerance ε (as tiny as you want). I must find a moment N after which every term stays within ε of the limit. Cauchy is the same idea without knowing the limit: the terms get close to each other.



$a_n = (-1)^n/n$ with $\varepsilon = 0.2$: from $N = \lceil 1/0.2 \rceil = 5$ on, every point sits inside the green tube. A smaller ε just needs a larger N.

THEOREM / PROPOSITION // Basic facts in $\mathbb{R}^d$

1. **Convergent $\iff$ Cauchy** (because $\mathbb{R}^d$ is *complete*; Exercise 9b).
2. **Coordinate-wise:** $x_n \to x^\star$ iff each coordinate sequence converges to the matching coordinate.
3. **Any norm works:** all norms on $\mathbb{R}^d$ are equivalent ($\alpha\|x\|_a \le \|x\|_b \le \beta\|x\|_a$), so the verdict "converges or not" never depends on the norm (Exercise 9c).
4. **Monotone convergence:** a real sequence that is increasing and bounded above converges to its supremum (decreasing + bounded below: to its infimum). Exercise 9a / 10a.
5. **Algebra of limits:** sums, scalar multiples, products, inner products and norms of convergent sequences converge to what you'd expect.

4.2 Proving convergence: build $N(\varepsilon)$

EXAM CHEAT CODE // "Show from the definition that $x_n \to L$"

1. **Scratch work:** compute $|x_n - L|$ and *bound it* by something simple that clearly shrinks, like $\frac{C}{n}$ or $C\rho^n$.
2. Solve "simple bound $< \varepsilon$" for n. That gives your $N(\varepsilon)$.
3. **Write it up:** "Let $\varepsilon > 0$. Take $N = \dots$. For $n \ge N$: $|x_n - L| \le \cdots \le \cdots < \varepsilon$."

EXAMPLE // Two ε–N proofs with numbers

(a) $\frac{(-1)^n}{n} \to 0$. $\left|\frac{(-1)^n}{n} - 0\right| = \frac{1}{n}$. We need $\frac{1}{n} < \varepsilon$, i.e. $n > \frac{1}{\varepsilon}$. Take $N = \lfloor 1/\varepsilon \rfloor + 1$.
Numbers: $\varepsilon = 0.05 \Rightarrow N = 21$; indeed $\frac{1}{21} \approx 0.048 < 0.05$. (Taking $N = \lceil 1/\varepsilon \rceil = 20$ also works if you only ask for $\le \varepsilon$.)
(b) $\rho^n \to 0$ **for** $0 < \rho < 1$**.** We need $\rho^n < \varepsilon$. Take logs: $n \log \rho < \log \varepsilon$. **Careful:** $\log \rho < 0$, so dividing flips the inequality: $n > \frac{\log \varepsilon}{\log \rho}$.
Numbers: $\rho = 0.7$, $\varepsilon = 0.05$: $\frac{\log 0.05}{\log 0.7} = \frac{-3.00}{-0.357} \approx 8.4$, so $N = 9$. Check: $0.7^9 \approx 0.040 < 0.05$, while $0.7^8 \approx 0.058$ is not. (Any larger N, like 23, is also valid, but 9 is the smallest one that works.)

COMMON L (don't do this) // Sequence traps

1. **Oscillation is not the problem; non-shrinking oscillation is.** $(-1)^n/n \to 0$, but $(-1)^n$ diverges.

2. **To show divergence**, find two subsequences with different limits: $(-1)^{2n} = 1$, $(-1)^{2n+1} = -1$.
3. **Bounded does not mean convergent** ($(-1)^n$). Monotone + bounded does.
4. **Monotone but unbounded** goes to $\pm\infty$ ($a_n = n$).
5. Unsure in $\mathbb{R}^d$? Check coordinates one by one.

ML SIDE QUEST // Sequences in training

- **Diagnostics:** watch $f(w_n)$ and the validation metric: stagnation, divergence or oscillation show up there first.
- **Stopping rules:** stop when $\|w_{n+1} - w_n\| < \text{tol}$ (a Cauchy-style test: you don't know the limit, but successive terms are close).
- **Running averages:** Adam/RMSProp use $\hat{g}_n = \alpha g_n + (1-\alpha)\hat{g}_{n-1}$ to smooth noisy gradients. That's a sequence defined by recursion, exactly like Exercise 2c.

5 Asymptotic orders: O, o, Θ

DEFINITION // Comparing growth as $n \to \infty$

- $x_n = O(y_n)$ ("at most of order y_n"): there are $C > 0$ and N with $|x_n| \le C|y_n|$ for $n \ge N$.
- $x_n = o(y_n)$ ("negligible compared to y_n"): for every $\varepsilon > 0$ there is N with $|x_n| \le \varepsilon|y_n|$ for $n \ge N$. Equivalently (when $y_n \neq 0$): $\dfrac{x_n}{y_n} \to 0$.
- $x_n = \Theta(y_n)$ ("exactly of order y_n"): there are $c_1, c_2 > 0$ and N with $c_1|y_n| \le |x_n| \le c_2|y_n|$ for $n \ge N$.

Same definitions for functions as $x \to a$ or $x \to \infty$.

Analogy: O is "$\le$", o is "$<$, and by a lot", Θ is "$\approx$, up to constants".

THEOREM / PROPOSITION // Rules

1. **Dominance ladder:** $\log n \ll n^\alpha \ll \beta^n$ for every $\alpha > 0$, $\beta > 1$ ("$\ll$" meaning o).
2. **Algebra:** $o(y_n) + o(y_n) = o(y_n)$, $O(a_n)\,O(b_n) = O(a_n b_n)$, $o(y_n) \subseteq O(y_n)$.
3. **Transitivity:** $x = O(y)$ and $y = O(z) \Rightarrow x = O(z)$ (same with o).
4. **Taylor language:** $f(x+h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$: the leftover is negligible compared to the step.

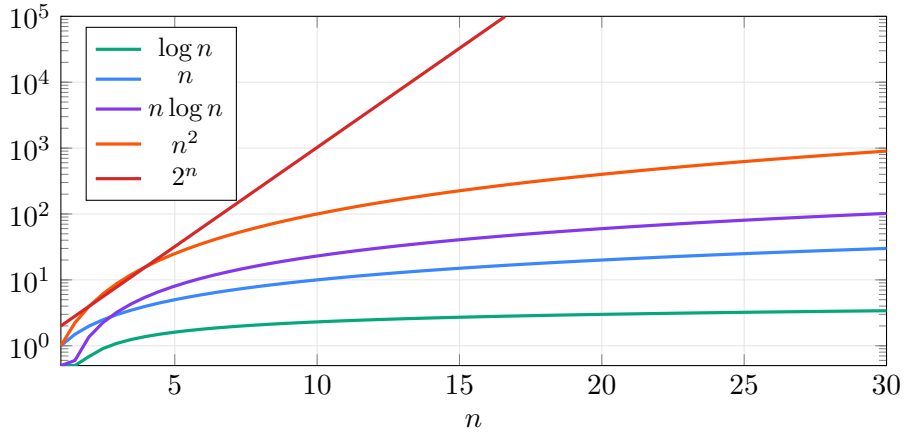
EXAM CHEAT CODE // "Compare x_n and y_n"

1. Compute $\lim \dfrac{x_n}{y_n}$.
2. Limit $= 0$: $x_n = o(y_n)$ (so also O).
3. Limit $= c \in]0, \infty[$: $x_n = \Theta(y_n)$.
4. Limit $= \infty$: $y_n = o(x_n)$.
5. No limit but the ratio stays bounded: O (and Θ if it also stays away from 0).

To find the limit: **factor out the biggest term**.

EXAMPLE // Worked comparisons

1. $n \log n$ vs n^2: $\frac{n \log n}{n^2} = \frac{\log n}{n} \to 0$, so $n \log n = o(n^2)$ (merge sort beats bubble sort).
2. $3n^2 + 5n$ vs n^2: ratio $3 + \frac{5}{n} \to 3$, so $\Theta(n^2)$. Lower-order terms don't matter.
3. $\log n$ vs $n^{0.1}$: ratio $\to 0$ (Exercise 2b style), so $\log n = o(n^{0.1})$, even though at $n = 1000$, $\log n \approx 6.9 > 2.0 \approx n^{0.1}$. Asymptotics are about *eventually*.
4. $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} = \Theta(\log n)$ (Exercise 2a).
5. n^{10} vs 2^n: $n^{10} = o(2^n)$, even though $n^{10} > 2^n$ until $n \approx 59$.



Log scale on the y-axis. Polynomials become gentle curves, the exponential becomes a straight line that leaves everyone behind.

ML SIDE QUEST // Rates you'll meet

- **Cost:** one SGD pass over n samples with d features is $O(nd)$; quasi-Newton methods can cost $O(nd^2)$ or more.
- **Convergence speed:** *linear* convergence $\|x_n - x^\star\| = O(\rho^n)$ vs *sublinear* $O(1/n)$. To reach error 10^{-6}: about $\frac{\log 10^{-6}}{\log \rho}$ steps for the first (e.g. 131 for $\rho = 0.9$), about 10^6 for the second.
- **Step-size schedules:** $\sum \eta_t = \infty$ (you can travel arbitrarily far) but $\sum \eta_t^2 < \infty$ (noise dies out); $\eta_t = 1/t$ does both.

COMMON L (don't do this) // O traps

1. O hides constants: $1000n$ is $O(n)$, and may be slower than n^2 for every n you'll ever see.
2. o needs the ratio to go to 0; bounded isn't enough.
3. Θ needs **both** bounds. $n = O(n^2)$ is true, $n = \Theta(n^2)$ is false.
4. "=" in $x_n = O(y_n)$ is a one-way street: $n = O(n^2)$ does not mean $n^2 = O(n)$.

6 Limits and continuity

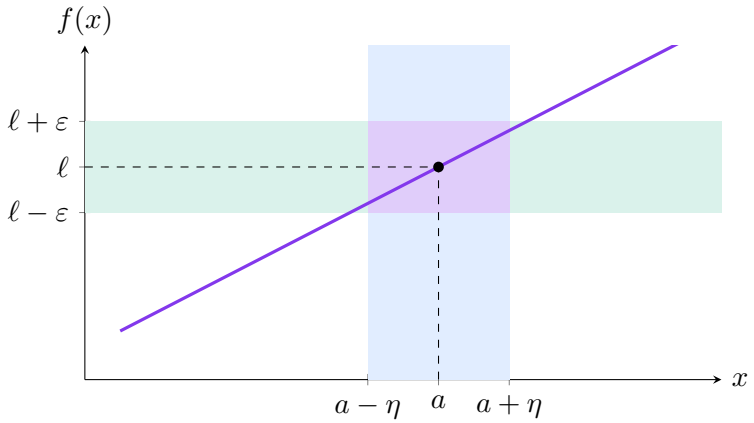
DEFINITION // ε–η limit, continuity

For $f : \mathbb{R} \to \mathbb{R}$: $\lim_{x \to a} f(x) = \ell$ iff for every $\varepsilon > 0$ there is $\eta > 0$ such that

$$0 < |x - a| < \eta \implies |f(x) - \ell| < \varepsilon.$$

f is **continuous at** a if $\lim_{x \to a} f(x) = f(a)$. In $\mathbb{R}^d$: replace $|\cdot|$ by a norm.

ε = how much output error you tolerate. η = how close to a the input must be. Same game as ε–N, with "x close to a" instead of "n large".



$f(x) = 3x + 1$ near $a = 2$, $\ell = 7$: inputs in the blue window land in the green band. Here $\varepsilon = 1.5$ and $\eta = \varepsilon/3 = 0.5$ (Exercise 11a).

THEOREM / PROPOSITION // Tools for computing limits

1. **Algebra:** add, subtract, multiply limits; divide only if the denominator's limit is $\neq 0$.
2. **Composition:** if $g(x) \to b$ and f is continuous at b, then $f(g(x)) \to f(b)$.
3. **Squeeze:** if $g \le f \le h$ near a and $g, h \to \ell$, then $f \to \ell$.
4. **One-sided limits:** $\lim_{x \to a} f$ exists iff $\lim_{x \to a^-} f$ and $\lim_{x \to a^+} f$ both exist and are equal.
5. **Sequential view:** f is continuous at a iff $x_n \to a$ implies $f(x_n) \to f(a)$ (Exercise 12b).

EXAM CHEAT CODE // "Compute $\lim_{x \to a} f(x)$"

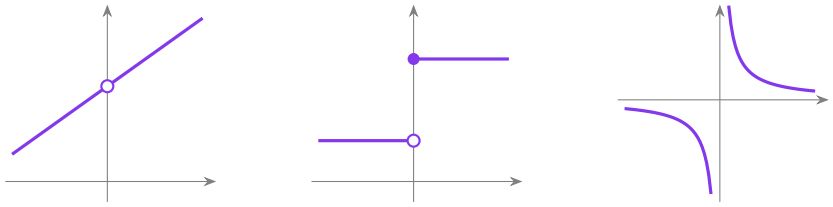
1. **Plug in** $x = a$. A real number with a non-zero denominator? Done.
2. Got $\frac{0}{0}$? It's **indeterminate**: factor and cancel, rationalise, use a known limit ($\frac{\sin x}{x} \to 1$), or squeeze.
3. Got $\frac{c}{0}$ with $c \neq 0$? **Blow-up**: check the sign on each side; no finite limit.
4. Piecewise? Compute the left and right limits separately.

EXAMPLE // The three kinds of trouble

1. **Removable:** $g(x) = \frac{x^2 - 1}{x - 1} = x + 1$ for $x \neq 1$, so $\lim_{x \to 1} g = 2$. There's a hole at 1; define $g(1) = 2$ and it's fixed.

2. **Jump:** $s(x) = -1$ for $x < 0$, 1 for $x \geq 0$. Left limit -1, right limit 1: no limit, no fix possible.
3. **Infinite:** $\frac{1}{x}$ at 0: $\to -\infty$ on the left, $+\infty$ on the right.

And the famous one: $\frac{\sin x}{x} \to 1$ (squeeze, Exercise 11b). Numerically: $\frac{\sin 0.1}{0.1} \approx 0.99833$.



removable jump infinite

ML SIDE QUEST // Continuity in ML

- **Activations** (ReLU, GELU, sigmoid) are continuous: small input changes give small output changes, which keeps training stable.
- **Near a good minimum**, continuity of the loss means tiny parameter nudges don't spike the loss.
- **Preprocessing:** patch artificial jumps (divide-by-zero, sentinel values like -999) so the model doesn't learn fake discontinuities.

COMMON L (don't do this) // Limit traps

1. $\frac{0}{0}$ is **not** 0 and not 1: it's "go do more work".
2. A two-sided limit needs both sides to agree.
3. Left limit = right limit is not enough for continuity: it must also equal $f(a)$ (Exercise 12c).
4. Continuous does **not** imply differentiable ($|x|$ at 0). Differentiable does imply continuous.

7 Derivative and first-order Taylor

DEFINITION // Derivative, linearisation

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

If it exists, f is **differentiable** at x, and

$$f(x+h) = f(x) + f'(x)\,h + o(|h|).$$

The **linearisation** (tangent line) at x_0 is $\ell_1(x) = f(x_0) + f'(x_0)(x - x_0)$.

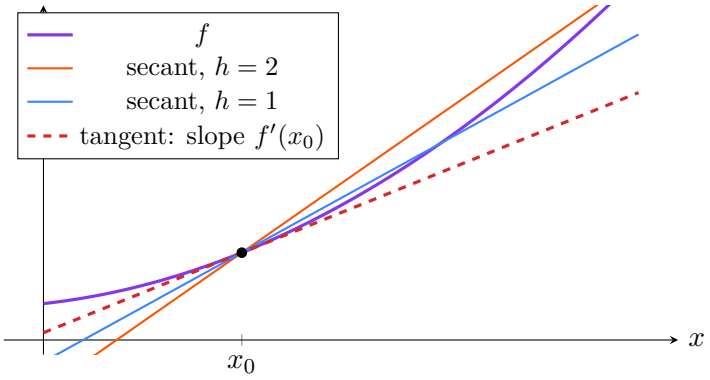
Zoom into a smooth curve enough and it looks like a straight line. $f'(x_0)$ is that line's slope; the $o(|h|)$ says the gap between curve and line shrinks faster than the step itself.

THEOREM / PROPOSITION // Differentiable ⇒ continuous

If f is differentiable at x, then f is continuous at x.

RECEIPTS (proof) //

$f(x+h) - f(x) = \frac{f(x+h)-f(x)}{h} \cdot h \;\to\; f'(x) \cdot 0 = 0$ as $h \to 0$. ■ *(and that's on periodt)*



Secant slopes $\frac{f(x_0+h)-f(x_0)}{h}$ for $h = 2$ and $h = 1$, approaching the tangent slope as $h \to 0$.

EXAM CHEAT CODE // "Linearise f at x_0 and check the error"

1. Compute $f(x_0)$.
2. Differentiate (sum, product, quotient, chain rules) and compute $f'(x_0)$.
3. Write $\ell_1(x) = f(x_0) + f'(x_0)(x - x_0)$.
4. Pick test points $x_0 \pm h$; compute $|f(x) - \ell_1(x)|$. It should be small for small h and grow roughly like h^2 (it's about $\frac{1}{2}|f''(x_0)|h^2$).

EXAMPLE // Worked example: $f(x) = e^{-0.2x^2} + 0.2\sin(2x)$ at $x_0 = 1$

1. $f(1) = e^{-0.2} + 0.2\sin 2 \approx 0.81873 + 0.18186 = 1.00059$.
2. Chain rule on each piece: $\left(e^{-0.2x^2}\right)' = -0.4x\, e^{-0.2x^2}$ and $\left(0.2\sin 2x\right)' = 0.4\cos 2x$. So

$$f'(x) = -0.4x\, e^{-0.2x^2} + 0.4\cos(2x), \qquad f'(1) \approx -0.32749 - 0.16646 = -0.49395.$$

3. $\ell_1(x) \approx 1.00059 - 0.49395\,(x - 1)$.
4. Check:

x	$f(x)$	$\ell_1(x)$	$\lvert f(x) - \ell_1(x)\rvert$
0.9	1.04521	1.04999	0.00477
1.1	0.94676	0.95120	0.00444
1.2	0.88485	0.90180	0.01695

Doubling h from 0.1 to 0.2 multiplies the error by about 4: the error is $\approx \frac{1}{2}|f''(1)|h^2$, with $f''(1) \approx -0.92$.

Contrast: $|x|$ is continuous at 0 but not differentiable: the difference quotient $\frac{|h|}{h}$ is -1 on the left and $+1$ on the right.

ML SIDE QUEST // Gradient descent in one line

Take a step $x \mapsto x - \eta f'(x)$ with $\eta > 0$. The linear model predicts

$$f(x - \eta f'(x)) \approx f(x) - \eta\,(f'(x))^2,$$

which is **smaller** than $f(x)$ whenever $f'(x) \neq 0$. That's the whole idea of gradient descent. The catch: the prediction is only trusted near x, so η must be small enough (Exercise 14). Also: **gradient checking** compares $\frac{f(x+h)-f(x-h)}{2h}$ with what autodiff returns, to catch bugs.

COMMON L (don't do this) // Derivative traps

1. Linearisations are **local**: the error grows as you move away.
2. Little-o vs big-O: $r(h) = o(|h|)$ means $r(h)/|h| \to 0$; $r(h) = O(|h|)$ only means bounded by $C|h|$.
3. At a kink, the left and right derivatives differ, so there's no derivative there.

8 Gradient and directional derivatives

DEFINITION // Partial derivatives, gradient

For $f : \mathbb{R}^n \to \mathbb{R}$, the i-th **partial derivative** is the ordinary derivative when you move only along axis i:

$$\frac{\partial f}{\partial x_i}(x) = \lim_{h \to 0} \frac{f(x + he_i) - f(x)}{h}.$$

The **gradient** stacks them: $\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \ldots, \frac{\partial f}{\partial x_n}(x)\right)^\top \in \mathbb{R}^n$.
f is **differentiable** at x if $f(x + h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$.

DEFINITION // Directional derivative

For a **unit** vector u: $D_u f(x) = \lim_{t \to 0} \frac{f(x + tu) - f(x)}{t} = \langle \nabla f(x), u\rangle$ (if f is differentiable).

THEOREM / PROPOSITION // The gradient points uphill, steepest

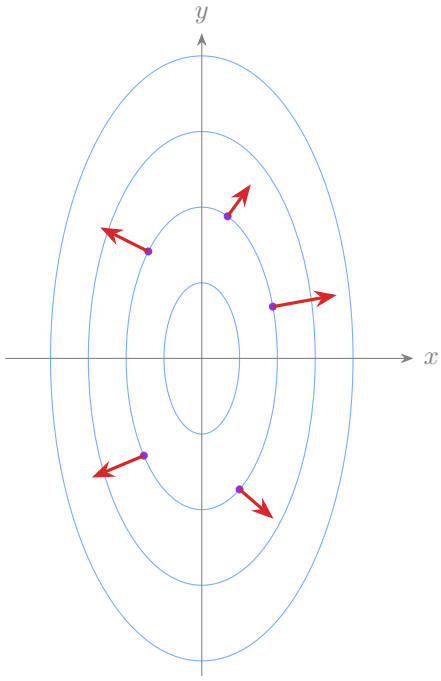
For differentiable f with $\nabla f(x) \neq 0$:

1. $\max_{\|u\|=1} D_u f(x) = \|\nabla f(x)\|$, reached at $u = \nabla f(x)/\|\nabla f(x)\|$: **steepest ascent**.
2. $-\nabla f(x)$ is the direction of **steepest descent**.
3. $\nabla f(x)$ is **orthogonal to the level set** $\{f = f(x)\}$ through x.

RECEIPTS (proof) // Cauchy–Schwarz, from Lecture 5

$D_u f(x) = \langle \nabla f(x), u\rangle \leq \|\nabla f(x)\|\|u\| = \|\nabla f(x)\|$, with equality iff u is a positive multiple of $\nabla f(x)$. Along a level curve f doesn't change, so the directional derivative along its tangent u is 0: $\langle \nabla f(x), u\rangle = 0$. ■ *(and that's on periodt)*

So $D_u f(x)$ is just the (signed) length of the shadow of $\nabla f(x)$ on u: the projection idea of Lecture 5.



level curves of $f(x,y) = x^2 + \frac{1}{4}y^2$ (ellipses $f = \text{const}$)

red arrows: $\nabla f = (2x, \frac{1}{2}y)$, perpendicular to the curves, pointing to higher values

EXAM CHEAT CODE // "Compute the gradient / a directional derivative"

1. For each variable, differentiate treating all the others as **constants**.
2. Stack them: $\nabla f = (\partial_1 f, \ldots, \partial_n f)$. Plug in the point.
3. **Normalise** the direction: $u = d/\|d\|$. (Forgetting this is the number one mistake.)
4. $D_u f = \langle \nabla f, u \rangle$. Max possible value: $\|\nabla f\|$.

EXAMPLE // A clean polynomial: $f(x,y) = x^2 y + 3y$ at $(1,2)$

1. $\frac{\partial f}{\partial x} = 2xy$ (y is a constant), $\frac{\partial f}{\partial y} = x^2 + 3$.
2. $\nabla f(1,2) = (4,4)$.
3. Direction $d = (3,4)$, $\|d\| = 5$, $u = (0.6, 0.8)$.
4. $D_u f = 4 \cdot 0.6 + 4 \cdot 0.8 = 5.6$. Max possible: $\|(4,4)\| = 4\sqrt{2} \approx 5.66$, at $u = \frac{1}{\sqrt{2}}(1,1)$.
5. Sanity check numerically: $\frac{f((1,2)+0.001u) - f(1,2)}{0.001} \approx 5.60$ ✓.

EXAMPLE // Worked example: $f(x,y) = x^2 + \frac{1}{4}y^2 + 0.5\sin(2x)\cos(2y)$ at $(0.6, -0.4)$

1. $\frac{\partial f}{\partial x} = 2x + 0.5 \cdot 2\cos(2x)\cos(2y) = 2x + \cos(2x)\cos(2y)$.
2. $\frac{\partial f}{\partial y} = \frac{1}{2}y + 0.5\sin(2x) \cdot (-2\sin(2y)) = \frac{1}{2}y - \sin(2x)\sin(2y)$.
3. At $(0.6, -0.4)$: $\cos 1.2 \approx 0.3624$, $\cos(-0.8) \approx 0.6967$, $\sin 1.2 \approx 0.9320$, $\sin(-0.8) \approx -0.7174$.
4. $\nabla f \approx (1.2 + 0.2525,\ -0.2 + 0.6686) = (1.452,\ 0.469)$.
5. $u = (1,0)$: $D_u f \approx 1.452$. $u = \frac{1}{\sqrt{2}}(1,1)$: $D_u f \approx \frac{1.452 + 0.469}{\sqrt{2}} \approx 1.358$. Steepest: $\|\nabla f\| \approx 1.526$.

ML SIDE QUEST // The gradient runs ML

- **Gradient descent:** $w_{k+1} = w_k - \eta \nabla f(w_k)$: step in the steepest-descent direction.
- **Saliency maps:** $\left|\frac{\partial \text{score}}{\partial x_i}\right|$ for each input pixel x_i shows which pixels the prediction is most sensitive to.
- **Projected gradient:** with constraints, take a gradient step, then project back onto the allowed set (Lecture 5 projections).

9 Jacobian and the differential

DEFINITION // Differential, Jacobian

$F : \mathbb{R}^n \to \mathbb{R}^m$ is **differentiable** at x if there is a linear map $DF(x) : \mathbb{R}^n \to \mathbb{R}^m$ with

$$F(x+h) = F(x) + DF(x)\,h + o(\|h\|).$$

Its matrix is the **Jacobian** $J_F(x) \in \mathbb{R}^{m \times n}$, with entries $[J_F(x)]_{ij} = \frac{\partial F_i}{\partial x_j}(x)$:

$$J_F(x) = \begin{pmatrix} \frac{\partial F_1}{\partial x_1} & \cdots & \frac{\partial F_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial x_1} & \cdots & \frac{\partial F_m}{\partial x_n} \end{pmatrix} \quad \leftarrow \text{ row } i = (\nabla F_i)^\top$$

Rows = one output each. **Columns** = one input direction each: column j is $J_F(x)e_j$, how a nudge along x_j moves all outputs. For $m = 1$ (scalar f): $J_f(x) = \nabla f(x)^\top$, a row.

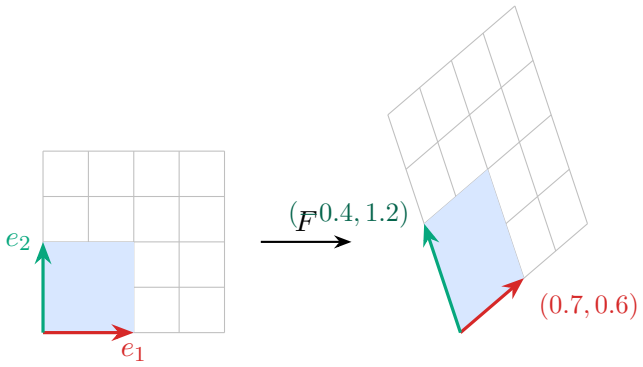
Shape rule: J_F is (number of outputs) $\times$ (number of inputs). If it isn't, you transposed something.

THEOREM / PROPOSITION // Linear maps are their own Jacobian

If $F(x) = Ax + b$, then $J_F(x) = A$ for every x (and the $o(\|h\|)$ term is exactly 0). So the Jacobian is "the matrix that F looks like up close".

EXAMPLE // Worked example: $F(x, y) = (0.7x - 0.4y,\ 0.6x + 1.2y)$

$J_F = \begin{pmatrix} 0.7 & -0.4 \\ 0.6 & 1.2 \end{pmatrix}$ everywhere. Columns: $e_1 \mapsto (0.7, 0.6)$, $e_2 \mapsto (-0.4, 1.2)$. The unit square becomes a parallelogram of area $|\det J_F| = |0.84 + 0.24| = 1.08$.



input grid image: columns of J_F, area $\times 1.08$

EXAM CHEAT CODE // "Compute the Jacobian and linearise F near x_0"

1. Write the components $F_1, \dots, F_m$.
2. Row i = gradient of F_i: $(\frac{\partial F_i}{\partial x_1}, \dots, \frac{\partial F_i}{\partial x_n})$.
3. **Check the shape**: $m \times n$.
4. Plug in x_0. Then $F(x_0 + h) \approx F(x_0) + J_F(x_0)\, h$.

EXAMPLE // A non-linear one: $F(x, y) = (x^2 y,\ x + \sin y)$ at $(1, 0)$

1. $J_F = \begin{pmatrix} 2xy & x^2 \\ 1 & \cos y \end{pmatrix}$, so $J_F(1,0) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Shape 2×2 ✓.
2. $F(1, 0) = (0, 1)$.
3. Linearise with $h = (0.1, 0.1)$: $F(1.1, 0.1) \approx (0, 1) + \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0.1 \\ 0.1 \end{pmatrix} = (0.1,\ 1.2)$.
4. True value: $(1.21 \cdot 0.1,\ 1.1 + \sin 0.1) = (0.121,\ 1.1998)$. Close ✓, error of order $\|h\|^2$.

ML SIDE QUEST // Jacobians in ML

- **Normalising flows:** a density pushed through an invertible F picks up a factor $|\det J_F|$; the training loss contains $\log|\det J_F(x)|$.
- **Multi-output models:** for logits $z = F(x) \in \mathbb{R}^K$, the row i of J_F says how class i's score reacts to each input feature.
- **Sensitivity:** $\|F(x + h) - F(x)\| \approx \|J_F(x)h\| \le \|J_F(x)\|_2 \|h\|$: the operator norm of Section 2 is back.
- **Autodiff** frameworks compute products with Jacobians without ever building the full matrix.

10 The chain rule (a.k.a. backprop)

THEOREM / PROPOSITION // Chain rule

If $G : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at x and $F : \mathbb{R}^m \to \mathbb{R}^p$ is differentiable at $G(x)$, then $F \circ G$ is differentiable at x and

$$D(F \circ G)(x) = DF(G(x)) \circ DG(x), \qquad \underbrace{J_{F\circ G}(x)}_{p\times n} = \underbrace{J_F(G(x))}_{p\times m}\ \underbrace{J_G(x)}_{m\times n}.$$

For $n = m = p = 1$ it's the usual $(f \circ g)'(x) = f'(g(x))\, g'(x)$.

Composition of functions becomes multiplication of their best linear approximations. The inner dimensions (m) must match, like any matrix product: that's your built-in error check.

RECEIPTS (proof) // why (idea)

$G(x + h) = G(x) + J_G h + o(\|h\|)$. Feed that into F, which near $G(x)$ looks like $F(G(x)) + J_F\, k$ for a small displacement k. With $k = J_G h + o(\|h\|)$: $F(G(x+h)) = F(G(x)) + J_F J_G\, h + o(\|h\|)$. The linear part is $J_F J_G$. ■ *(and that's on periodt)*

EXAM CHEAT CODE // "Differentiate a composition"

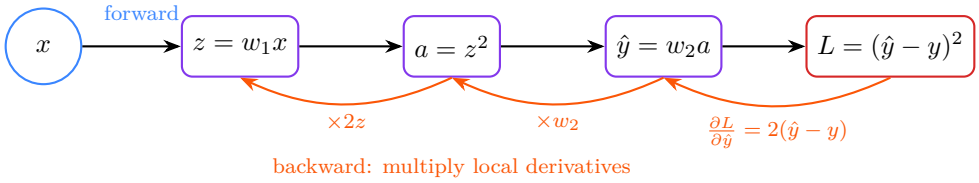
1. Name the layers: $x \xrightarrow{G} z \xrightarrow{F} y$.
2. Compute each Jacobian separately: $J_G(x)$ and $J_F(z)$.
3. Write the shapes. They must chain: $(p \times m)(m \times n)$.
4. Evaluate J_F **at** $z = G(x)$, not at x.
5. Multiply. For a scalar output, the gradient is the transpose: $\nabla(f \circ G)(x) = J_G(x)^\top \nabla f(G(x))$.

EXAMPLE // Worked example: $g(x) = (\sin x_1 + x_2^2,\ x_1 e^{x_2})$, $f(z) = z_1^2 + 3z_2$

1. Shapes: $g : \mathbb{R}^2 \to \mathbb{R}^2$, $f : \mathbb{R}^2 \to \mathbb{R}$. So J_g is 2×2, $J_f = \nabla f^\top$ is 1×2, result 1×2.
2. $J_g(x) = \begin{pmatrix} \cos x_1 & 2x_2 \\ e^{x_2} & x_1 e^{x_2} \end{pmatrix}$, $\quad \nabla f(z) = (2z_1,\ 3)$.
3. Multiply: $J_{f\circ g}(x) = (2z_1,\ 3) \begin{pmatrix} \cos x_1 & 2x_2 \\ e^{x_2} & x_1 e^{x_2} \end{pmatrix} = (2z_1 \cos x_1 + 3e^{x_2},\ \ 4z_1 x_2 + 3x_1 e^{x_2})$ with $z_1 = \sin x_1 + x_2^2$.
4. **Numbers at** $x = (0, 1)$**:** $z = g(0, 1) = (1, 0)$, $J_g = \begin{pmatrix} 1 & 2 \\ e & 0 \end{pmatrix}$, $\nabla f(z) = (2, 3)$.

$$(2, 3) \begin{pmatrix} 1 & 2 \\ e & 0 \end{pmatrix} = (2 + 3e,\ 4) \approx (10.15,\ 4).$$

5. **Check directly:** $f(g(x)) = (\sin x_1 + x_2^2)^2 + 3x_1 e^{x_2}$. Then $\partial_{x_1} = 2(\sin x_1 + x_2^2)\cos x_1 + 3e^{x_2} = 2 + 3e$ and $\partial_{x_2} = 4x_2(\sin x_1 + x_2^2) + 3x_1 e^{x_2} = 4$ at $(0, 1)$ ✓.



EXAMPLE // Backprop by hand on a tiny network

$x = 2$, $w_1 = 0.5$, $w_2 = 3$, target $y = 1$. Graph above.

1. **Forward:** $z = w_1 x = 1$, $a = z^2 = 1$, $\hat{y} = w_2 a = 3$, $L = (3 - 1)^2 = 4$.
2. **Backward, one node at a time:**
   - $\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y) = 4$;
   - $\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot a = 4 \cdot 1 = 4$;
   - $\frac{\partial L}{\partial a} = \frac{\partial L}{\partial \hat{y}} \cdot w_2 = 12$;
   - $\frac{\partial L}{\partial z} = \frac{\partial L}{\partial a} \cdot 2z = 24$;
   - $\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z} \cdot x = 48$.
3. **Check:** $L(w_1) = (3(2w_1)^2 - 1)^2 = (12w_1^2 - 1)^2$, so $L'(w_1) = 2(12w_1^2 - 1) \cdot 24 w_1 = 2 \cdot 2 \cdot 12 = 48$ at $w_1 = 0.5$ ✓.

Each backward step reuses the previous result: that's why backprop is cheap.

EXAMPLE // Chain rule with matrices: the least-squares gradient

$f(w) = \|Xw - y\|_2^2$ with $X \in \mathbb{R}^{n\times d}$. Layers: $w \xrightarrow{G} r = Xw - y \xrightarrow{F} \|r\|^2$.

1. $J_G = X$ $(n \times d)$, $\nabla F(r) = 2r$, so $J_F = 2r^\top$ $(1 \times n)$.
2. $J_f = 2r^\top X$ $(1 \times d)$ ✓, so $\nabla f(w) = 2X^\top(Xw - y)$.
3. Setting it to 0 gives the normal equations $X^\top Xw = X^\top y$ (Lecture 2).

ML SIDE QUEST // Vanishing and exploding gradients

For a deep network $f = f_L \circ \cdots \circ f_1$, the chain rule gives $J_f = J_{f_L} \cdots J_{f_1}$, so $\|J_f\|_2 \le \prod_\ell \|J_{f_\ell}\|_2$ (submultiplicativity, Section 2). Factors a bit above 1, multiplied 50 times: $1.1^{50} \approx 117$ (explode). A bit below 1: $0.9^{50} \approx 0.005$ (vanish). That's why initialisation, normalisation and residual connections try to keep each layer's Jacobian norm near 1.

11 The exercises, fully solved

TL;DR // How to use this section

Try first, then check. Each solution starts with the **move** it uses. Numbers were re-checked by computer. Three classic traps are flagged where they happen (Exercises 7, 13d and 14b).

11.1 Sequences and asymptotics (Exercises 1, 2, 8, 9, 10)

SOLUTION // Exercise 1a: $a_n = \frac{(-1)^n}{n} \to 0$ from the definition

Move: bound $|a_n - 0|$, then solve for n.

1. $|a_n - 0| = \frac{|(-1)^n|}{n} = \frac{1}{n}$.
2. Let $\varepsilon > 0$. Choose an integer $N > \frac{1}{\varepsilon}$ (Archimedean property).
3. For $n \ge N$: $|a_n - 0| = \frac{1}{n} \le \frac{1}{N} < \varepsilon$. ■ *(and that's on periodt)*

E.g. $\varepsilon = 0.01 \Rightarrow N = 101$.

SOLUTION // Exercise 1b: $v_n = (\frac{1}{n}, \frac{(-1)^n}{n}, 1 - \frac{1}{n}) \to (0, 0, 1)$

Move: in $\mathbb{R}^d$, convergence = convergence of each coordinate.

1. $\frac{1}{n} \to 0$; $\frac{(-1)^n}{n} \to 0$ (part a); $1 - \frac{1}{n} \to 1$.
2. So $v_n \to (0, 0, 1)$. Directly with the max-norm: $\|v_n - (0,0,1)\|_\infty = \max(\frac{1}{n}, \frac{1}{n}, \frac{1}{n}) = \frac{1}{n} \to 0$. Any other norm gives the same verdict (norm equivalence).

SOLUTION // Exercise 2a: $H_n = \sum_{k=1}^n \frac{1}{k} = \Theta(\log n)$

Move: compare the sum with an integral ($t \mapsto \frac{1}{t}$ is decreasing).

1. For $t \in [k, k+1]$: $\frac{1}{k+1} \le \frac{1}{t} \le \frac{1}{k}$. Integrate over $[k, k+1]$: $\frac{1}{k+1} \le \int_k^{k+1} \frac{dt}{t} \le \frac{1}{k}$.

2. **Lower bound:** sum the right inequality for $k = 1..n$: $\log(n+1) = \int_1^{n+1} \frac{dt}{t} \le H_n$.
3. **Upper bound:** sum the left one for $k = 1..n-1$: $H_n - 1 \le \int_1^n \frac{dt}{t} = \log n$, so $H_n \le 1 + \log n$.
4. For $n \ge 3$: $\log n \ge 1$, so $\log n \le \log(n+1) \le H_n \le 1 + \log n \le 2\log n$. That's $\Theta(\log n)$ with $c_1 = 1$, $c_2 = 2$.

Numbers: $H_{100} \approx 5.187$, $\log 100 \approx 4.605$, $1 + \log 100 \approx 5.605$ ✓.

SOLUTION // Exercise 2b: $n^\alpha = o(\beta^n)$ for $\alpha > 0$, $\beta > 1$

Move: ratio of consecutive terms. Let $u_n = \frac{n^\alpha}{\beta^n} > 0$.

1. $\frac{u_{n+1}}{u_n} = (1 + \frac{1}{n})^\alpha \cdot \frac{1}{\beta} \to \frac{1}{\beta} < 1$.
2. Pick q with $\frac{1}{\beta} < q < 1$. There's N with $\frac{u_{n+1}}{u_n} \le q$ for $n \ge N$.
3. Then $u_n \le u_N q^{n-N}$ for $n \ge N$, and $q^{n-N} \to 0$. So $u_n \to 0$, i.e. $n^\alpha = o(\beta^n)$. ■ *(and that's on periodt)*

Alternative: $u_n = \exp(\alpha \log n - n \log \beta) = \exp(-n(\log \beta - \alpha \frac{\log n}{n})) \to 0$ since $\frac{\log n}{n} \to 0$.

SOLUTION // Exercise 2c: rate of $x_{k+1} = \rho x_k + \frac{c}{k+1}$, $0 < \rho < 1$, $c > 0$

Move: unroll the recursion, then bound the sum.

1. Unroll: $x_1 = \rho x_0 + c$, $x_2 = \rho^2 x_0 + \rho c + \frac{c}{2}$, …, in general

$$x_k = \rho^k x_0 + c\, S_k, \qquad S_k = \sum_{\ell=0}^{k-1} \frac{\rho^\ell}{k-\ell}.$$

2. **Lower bound:** keep only $\ell = 0$: $S_k \ge \frac{1}{k}$.
3. **Upper bound:** split at $\ell = k/2$.
   - $\ell \le k/2$: $k - \ell \ge k/2$, so these terms sum to at most $\frac{2}{k} \sum_\ell \rho^\ell \le \frac{2}{k(1-\rho)}$.
   - $\ell > k/2$: $k - \ell \ge 1$, so these terms sum to at most $\sum_{\ell > k/2} \rho^\ell \le \frac{\rho^{k/2}}{1-\rho}$, which is $o(1/k)$ (exponential beats $1/k$, Exercise 2b).
4. So $\frac{c}{k} \le cS_k \le \frac{2c}{(1-\rho)k} + o(\frac{1}{k})$, i.e. $cS_k = \Theta(1/k)$. And $\rho^k x_0 = o(1/k)$.
5. **Conclusion:** $x_k = \Theta(1/k)$. The geometric part dies fast; the $\frac{c}{k+1}$ "noise" sets the slow final rate.

Bonus (exact constant): in fact $k\, x_k \to \frac{c}{1-\rho}$. Heuristic: plug $x_k \approx \frac{a}{k}$ into the recursion, $\frac{a}{k+1} \approx \rho \frac{a}{k} + \frac{c}{k+1}$, so $a(1-\rho) = c$. Numerically with $\rho = 0.5$, $c = 1$: $1000\, x_{1000} \approx 2.002$, and $\frac{c}{1-\rho} = 2$ ✓.

SOLUTION // Exercise 8: limits of four sequences

Move: factor out the dominant term.

a) $a_n = \dfrac{3n^2 + 5n}{2n^2 - 1} = \dfrac{3 + 5/n}{2 - 1/n^2} \to \dfrac{3}{2}$.

b) $b_n = \dfrac{2^n + n^{10}}{3 \cdot 2^n - 5} = \dfrac{1 + n^{10}/2^n}{3 - 5/2^n} \to \dfrac{1+0}{3-0} = \dfrac{1}{3}$, using $n^{10} = o(2^n)$ (Exercise 2b). *Don't trust small n: at $n = 10$, $b_n \approx 3.3$ million, since 10^{10} dwarfs $2^{10} = 1024$!*

c) $c_n = \sin(n\pi) + \frac{1}{n}$: $\sin(n\pi) = 0$ for every integer n, so $c_n = \frac{1}{n} \to 0$.

d) $d_n = (-0.9)^n$: $|d_n| = 0.9^n \to 0$, so $d_n \to 0$ (it oscillates, but the amplitude shrinks).

SOLUTION // Exercise 9a (and 10a): bounded increasing ⇒ convergent

Move: the limit is the supremum.

1. The set $\{a_n\}$ is non-empty and bounded above, so $s = \sup_n a_n$ exists (completeness of $\mathbb{R}$).
2. Let $\varepsilon > 0$. Since $s - \varepsilon$ is not an upper bound, there's N with $a_N > s - \varepsilon$.
3. For $n \geq N$: $s - \varepsilon < a_N \leq a_n \leq s$ (increasing, and s is an upper bound). So $|a_n - s| < \varepsilon$. ■ *(and that's on periodt)*

Decreasing and bounded below: apply this to $(-a_n)$; the limit is $\inf_n a_n$. **For functions (10a):** a monotone bounded f on $[a, \infty[$ has a limit at $+\infty$, and a monotone f has one-sided limits at every point; same sup/inf proof.

SOLUTION // Exercise 9b: in $\mathbb{R}^d$, Cauchy $\iff$ convergent

($\Leftarrow$) Move: triangle inequality. If $x_n \to x$: given ε, take N with $\|x_n - x\| < \varepsilon/2$ for $n \geq N$. Then $\|x_n - x_m\| \leq \|x_n - x\| + \|x - x_m\| < \varepsilon$.
($\Rightarrow$) Move: bounded → convergent subsequence → whole sequence.

1. **Cauchy ⇒ bounded:** with $\varepsilon = 1$, for $n \geq N$, $\|x_n\| \leq \|x_N\| + 1$; finitely many earlier terms are bounded too.
2. **Bolzano–Weierstrass** (Exercise 10c), coordinate by coordinate: a bounded sequence in $\mathbb{R}^d$ has a convergent subsequence $x_{n_k} \to x$. (Extract for coordinate 1, then from that subsequence for coordinate 2, etc.)
3. **Whole sequence:** given ε, take N from the Cauchy property for $\varepsilon/2$, and k with $n_k \geq N$ and $\|x_{n_k} - x\| < \varepsilon/2$. For $n \geq N$: $\|x_n - x\| \leq \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \varepsilon$. ■ *(and that's on periodt)*

SOLUTION // Exercise 9c: coordinate-wise convergence ⇒ convergence in every norm

Move: compare everything with $\|\cdot\|_\infty$.

1. Coordinate-wise convergence means $\|x_n - x\|_\infty = \max_i |x_{n,i} - x_i| \to 0$ (a max of finitely many sequences going to 0).
2. For any norm $\|\cdot\|_\alpha$: write $v = \sum_i v_i e_i$, so $\|v\|_\alpha \leq \sum_i |v_i| \|e_i\|_\alpha \leq C_\alpha \|v\|_\infty$ with $C_\alpha = \sum_i \|e_i\|_\alpha$.
3. So $\|x_n - x\|_\alpha \leq C_\alpha \|x_n - x\|_\infty \to 0$. Same for $\|\cdot\|_\beta$. ■ *(and that's on periodt)*

SOLUTION // Exercise 10b: $\sum_{k\geq 1} \frac{1}{k^p}$ converges iff $p > 1$

Move: integral test.

1. If $p \leq 0$: $\frac{1}{k^p} \geq 1$ doesn't go to 0, so the series diverges.
2. If $p > 0$: $t \mapsto t^{-p}$ is positive and decreasing, so as in Exercise 2a:

$$\int_1^{N+1} t^{-p}\, dt \;\leq\; \sum_{k=1}^{N} k^{-p} \;\leq\; 1 + \int_1^{N} t^{-p}\, dt.$$

3. $p > 1$: $\int_1^N t^{-p} dt = \frac{1 - N^{1-p}}{p-1} \leq \frac{1}{p-1}$. Partial sums are increasing and bounded by $1 + \frac{1}{p-1}$, so they converge (monotone convergence).

4. $p = 1$: $\int_1^{N+1} \frac{dt}{t} = \log(N+1) \to \infty$. Diverges.
5. $0 < p < 1$: $\int_1^{N+1} t^{-p} dt = \frac{(N+1)^{1-p}-1}{1-p} \to \infty$. Diverges. ■ *(and that's on periodt)*

Example: $\sum \frac{1}{k^2} = \frac{\pi^2}{6} \approx 1.645 \leq 1 + \frac{1}{2-1} = 2$ ✓.

SOLUTION // Exercise 10c: Bolzano–Weierstrass in $\mathbb{R}$

Move: every sequence has a monotone subsequence, then use monotone convergence.

1. Call n a *peak* if $a_n \geq a_m$ for all $m > n$ (nothing later is higher).
2. **Infinitely many peaks** $n_1 < n_2 < \ldots$: then $a_{n_1} \geq a_{n_2} \geq \ldots$, a decreasing subsequence.
3. **Finitely many peaks:** after the last one, start at some n_1; it isn't a peak, so there's $n_2 > n_1$ with $a_{n_2} > a_{n_1}$; n_2 isn't a peak either, so there's $n_3 > n_2$ with $a_{n_3} > a_{n_2}$, and so on: an increasing subsequence.
4. Either way the subsequence is monotone, and bounded (since (a_n) is), so it converges (Exercise 9a). ■ *(and that's on periodt)*

11.2 Operator norms and SVD (Exercises 3–7)

SOLUTION // Exercise 3: $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

a) Column sums $2+1=3$, $1+2=3$: $\|A\|_1 = 3$. Row sums 3, 3: $\|A\|_\infty = 3$.

b) $A^\top A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\chi(\lambda) = (5-\lambda)^2 - 16 = 0 \iff \lambda = 9$ or 1. So $\|A\|_2 = \sqrt{9} = 3$.

c) $v = \frac{1}{\sqrt{2}}(1,1)$ (eigenvector of $A^\top A$ for 9). $Av = \frac{1}{\sqrt{2}}(3,3) = 3v$: same direction, stretched by exactly 3. Here all three norms equal 3, but that's a coincidence of this symmetric matrix.

SOLUTION // Exercise 4: sphere vs ball, and the "largest radius"

a) $\sup_{\|x\|=1}\|Ax\| = \sup_{\|x\|\leq 1}\|Ax\|$**.** Call them S and B.

1. The sphere is inside the ball, so $S \leq B$.
2. Take $\|x\| \leq 1$. If $x = 0$, $\|Ax\| = 0 \leq S$. Otherwise $u = x/\|x\|$ is a unit vector and $\|Ax\| = \|x\|\,\|Au\| \leq 1 \cdot S$. So $B \leq S$.

b) Largest radius. $A(S^{n-1}) = \{Ax : \|x\| = 1\}$. Its largest radius is $\sup\{\|y\| : y \in A(S^{n-1})\} = \sup_{\|x\|=1}\|Ax\| = \|A\|_{\text{op}}$, by definition. It's actually a **max**: the sphere is closed and bounded (compact) and $x \mapsto \|Ax\|$ is continuous, so the sup is attained.

SOLUTION // Exercise 5: normal A ($AA^\top = A^\top A$) $\Rightarrow \|A\|_2 = \max_i |\lambda_i|$

Move: normal matrices diagonalise with orthonormal (complex) eigenvectors.

1. **Fact** (spectral theorem for normal matrices, the complex version of Lecture 5): $A = UDU^*$ with U unitary ($U^*U = I$, $U^* = \bar{U}^\top$) and $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$, $\lambda_i \in \mathbb{C}$.
2. $A^\top A = A^*A = UD^*U^*UDU^* = U\,\text{diag}(|\lambda_1|^2, \ldots, |\lambda_n|^2)\,U^*$.
3. So the eigenvalues of $A^\top A$ are $|\lambda_i|^2$, the singular values are $|\lambda_i|$, and $\|A\|_2 = \sigma_{\max} =$

$\max_i |\lambda_i|$. ■ *(and that's on periodt)*

Real symmetric case (no complex numbers needed): $A = VDV^\top \Rightarrow A^\top A = VD^2V^\top$, eigenvalues λ_i^2.

Examples: rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is normal, eigenvalues $\pm i$, and $\|A\|_2 = 1 = |\pm i|$ ✓. **Counterexample when not normal:** $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $\max|\lambda| = 0$ but $\|N\|_2 = 1$. So "spectral norm = biggest eigenvalue" is **false** in general.

SOLUTION // Exercise 6a: Eckart–Young in spectral norm

Let $A_k = \sum_{i \le k} \sigma_i u_i v_i^\top$.

Step 1: the error of A_k. $A - A_k = \sum_{i>k} \sigma_i u_i v_i^\top = U \operatorname{diag}(0, \dots, 0, \sigma_{k+1}, \dots, \sigma_r) V^\top$. That's an SVD, so $\|A - A_k\|_2 =$ its largest singular value $= \sigma_{k+1}$.

Step 2: nobody does better. Take any B with $\operatorname{rank} B \le k$.

1. By rank–nullity, $\dim \ker B \ge n - k$.
2. $W = \operatorname{span}(v_1, \dots, v_{k+1})$ has dimension $k+1$.
3. $(n-k) + (k+1) > n$, so $\ker B \cap W \neq \{0\}$: pick a unit x in both.
4. Write $x = \sum_{i \le k+1} c_i v_i$ with $\sum c_i^2 = 1$. Since $Bx = 0$:

$$\|(A-B)x\|_2^2 = \|Ax\|_2^2 = \Big\| \sum_{i \le k+1} c_i \sigma_i u_i \Big\|^2 = \sum_{i \le k+1} c_i^2 \sigma_i^2 \ge \sigma_{k+1}^2 \sum c_i^2 = \sigma_{k+1}^2.$$

5. So $\|A - B\|_2 \ge \sigma_{k+1} = \|A - A_k\|_2$. ■ *(and that's on periodt)*

SOLUTION // Exercise 6b: Eckart–Young in Frobenius norm

Step 1: the error of A_k. Frobenius norm is orthogonally invariant ($\|QMR\|_F = \|M\|_F$), so $\|A - A_k\|_F^2 = \|\operatorname{diag}(0, \dots, 0, \sigma_{k+1}, \dots, \sigma_r)\|_F^2 = \sum_{i>k} \sigma_i^2$.

Step 2: nobody does better. It's tempting to say "a similar argument"; here's a real one. We use **Weyl's inequality** for singular values:

$$\sigma_{i+j-1}(X+Y) \le \sigma_i(X) + \sigma_j(Y).$$

(Proof idea: same kernel-intersection trick as 6a, applied to the best rank-$(i-1)$ approximation of X and rank-$(j-1)$ of Y; together they have rank $\le i + j - 2$, so by 6a $\sigma_{i+j-1}(X+Y) \le \|X + Y - X_{i-1} - Y_{j-1}\|_2 \le \sigma_i(X) + \sigma_j(Y)$.)

1. Let $\operatorname{rank} B \le k$. Apply Weyl with $X = A - B$, $Y = B$, $j = k+1$: since $\sigma_{k+1}(B) = 0$,

$$\sigma_{i+k}(A) \le \sigma_i(A - B) \quad \text{for every } i \ge 1.$$

2. Square and sum over i: $\|A - B\|_F^2 = \sum_i \sigma_i(A-B)^2 \ge \sum_{i \ge 1} \sigma_{i+k}(A)^2 = \sum_{i>k} \sigma_i^2 = \|A - A_k\|_F^2$. ■ *(and that's on periodt)*

Picture: spectral error = only the first discarded σ; Frobenius error = all discarded σ's, Pythagoras-style.

SOLUTION // Exercise 7: low-rank regression $\min_{\operatorname{rank} W \le k} \|XW - Y\|_F$

Setup: $X = P\Sigma Q^\top$ (thin SVD), $P \in \mathbb{R}^{n\times d}$ with $P^\top P = I_d$, $\Sigma \in \mathbb{R}^{d\times d}$ invertible, Q orthogonal.
a) Reduce the problem.

1. Set $Z = Q^\top W$, i.e. $W = QZ$. It's a bijection, and $\operatorname{rank} Z = \operatorname{rank} W$ (Q invertible).
2. $XW = P\Sigma Q^\top QZ = P\Sigma Z$.
3. Split Y into its part in range P and the rest: $Y = PP^\top Y + (I - PP^\top)Y$. Then $P\Sigma Z - Y = P(\Sigma Z - P^\top Y) - (I - PP^\top)Y$, and these two pieces are orthogonal (one has columns in range P, the other in $(\operatorname{range} P)^\perp$). Pythagoras:
$$\|XW - Y\|_F^2 = \|\Sigma Z - P^\top Y\|_F^2 + \underbrace{\|(I - PP^\top)Y\|_F^2}_{\text{does not depend on } Z}.$$
4. So minimising over W is the same as minimising $\|\Sigma Z - P^\top Y\|_F$ over $\operatorname{rank} Z \le k$ (same minimisers).

Heads-up: it's tempting to write $\|P\Sigma Z - Y\|_F = \|\Sigma Z - P^\top Y\|_F$. That's only true if $n = d$. In general the two differ by the constant above. Same minimiser, different value.
b) Solve it. Let $B = P^\top Y$ and $M = \Sigma Z$. Since Σ is invertible, $\operatorname{rank} M = \operatorname{rank} Z$, and $Z \mapsto M$ is a bijection. So the problem is $\min_{\operatorname{rank} M \le k} \|M - B\|_F$, whose answer is $M^\star = [B]_k$ (Eckart–Young, 6b). Hence $Z^\star = \Sigma^{-1}[P^\top Y]_k$.
c) Back to W. $W^\star = QZ^\star = Q\Sigma^{-1}[P^\top Y]_k$, and
$$\|XW^\star - Y\|_F^2 = \sum_{i>k} \sigma_i(P^\top Y)^2 + \|(I - PP^\top)Y\|_F^2.$$

Interpretation: the second term is the error *any* linear model makes (the part of Y that X can't explain). The first is the price of the rank constraint: small if the singular values of $P^\top Y$ drop fast after the k-th one. Practical recipe: SVD of X, form $P^\top Y$, SVD of that, truncate, map back.

11.3 Limits, continuity, derivatives (Exercises 11–14)

SOLUTION // Exercise 11a: $\lim_{x\to 2}(3x+1) = 7$ with ε–η

1. Scratch: $|(3x+1) - 7| = |3x - 6| = 3|x - 2|$.
2. Let $\varepsilon > 0$, take $\eta = \varepsilon/3$.
3. If $|x - 2| < \eta$: $|(3x+1) - 7| = 3|x - 2| < 3\eta = \varepsilon$. ■ *(and that's on periodt)*

E.g. $\varepsilon = 0.3 \Rightarrow \eta = 0.1$: $x \in]1.9, 2.1[$ gives $3x + 1 \in]6.7, 7.3[$ ✓.

SOLUTION // Exercise 11b: $\lim_{x\to 0} \frac{\sin x}{x} = 1$ (squeeze)

1. For $0 < x < \frac{\pi}{2}$, comparing areas in the unit circle (triangle $\le$ sector $\le$ big triangle): $\sin x \le x \le \tan x$.
2. Divide by $\sin x > 0$: $1 \le \frac{x}{\sin x} \le \frac{1}{\cos x}$. Flip: $\cos x \le \frac{\sin x}{x} \le 1$.
3. Both sides are even functions, so this holds for $0 < |x| < \frac{\pi}{2}$ too.
4. $\cos x \to 1$ and $1 \to 1$: by squeeze, $\frac{\sin x}{x} \to 1$. ■ *(and that's on periodt)*

SOLUTION // Exercise 11c: $h(x) = x^2$ $(x < 1)$, $2x$ $(x \geq 1)$

1. $\lim_{x\to 1^-} h(x) = \lim x^2 = 1$.
2. $\lim_{x\to 1^+} h(x) = \lim 2x = 2$.
3. $1 \neq 2$: **no two-sided limit**, so h is **not continuous** at 1 (jump of size 1). It is right-continuous: $h(1) = 2 =$ right limit.

SOLUTION // Exercise 11d: classify the discontinuities

1. $g(x) = \frac{x^2-1}{x-1} = \frac{(x-1)(x+1)}{x-1} = x + 1$ for $x \neq 1$. So $\lim_{x\to 1} g = 2$ exists but $g(1)$ is undefined: **removable**. Setting $g(1) = 2$ makes it continuous.
2. $s(x) = -1$ $(x < 0)$, 1 $(x \geq 0)$: left limit -1, right limit 1: **jump**. No value of $s(0)$ can fix it.

SOLUTION // Exercise 12a: indeterminate or not?

(i) $\frac{x^2-4}{x-2}$ at 2: plugging in gives $\frac{0}{0}$, **indeterminate**. Factor: $= x + 2 \to 4$.

(ii) $\frac{\sin x}{x^2}$ at 0: $\frac{0}{0}$, **indeterminate**. Write $\frac{\sin x}{x} \cdot \frac{1}{x}$: the first factor $\to 1$, the second blows up. So $\to +\infty$ from the right, $-\infty$ from the left: **no limit**. (Indeterminate means "more work needed", not "a limit exists".)

(iii) $\frac{2}{x-1}$ at 1: $\frac{2}{0}$, **not** indeterminate: it's a blow-up. $+\infty$ from the right, $-\infty$ from the left, no limit.

SOLUTION // Exercise 12b: continuity ⇒ sequential continuity

Move: chain the two definitions. Let $x_n \to a$ and f continuous at a. Fix $\varepsilon > 0$.

1. Continuity: there's $\eta > 0$ with $|x - a| < \eta \Rightarrow |f(x) - f(a)| < \varepsilon$.
2. Convergence, used with tolerance η: there's N with $n \geq N \Rightarrow |x_n - a| < \eta$.
3. So $n \geq N \Rightarrow |f(x_n) - f(a)| < \varepsilon$, i.e. $f(x_n) \to f(a)$. ■ *(and that's on periodt)*

(The converse also holds, which is why you can test continuity with sequences.)

SOLUTION // Exercise 12c: equal one-sided limits are not enough

$f(x) = 0$ for $x \neq 0$ and $f(0) = 1$. Then $\lim_{0^-} f = \lim_{0^+} f = 0$, the limit exists and equals 0, but $f(0) = 1 \neq 0$: **not continuous**. (A removable discontinuity again: redefine $f(0) = 0$.)

SOLUTION // Exercise 13: linearising $f(x) = e^{-0.2x^2} + 0.2\sin(2x)$ at $x_0 = 1$

a) Chain rule on each term: $f'(x) = -0.4x\, e^{-0.2x^2} + 0.4\cos(2x)$.
b) $f(1) \approx 1.00059$, $f'(1) \approx -0.49395$, so $\ell_1(x) \approx 1.00059 - 0.49395(x - 1)$.
c)

x	$f(x)$	$\ell_1(x)$	$\lvert f - \ell_1 \rvert$
0.9	1.04521	1.04999	0.00477
1.1	0.94676	0.95120	0.00444

Both errors are below 0.5% of $f(1)$.
d) Relative error $E(h) = \frac{|f(1+h) - \ell_1(1+h)|}{|f(1)|}$. Test both sides:

$\lvert h\rvert$	0.2	0.3	0.4
$E(+\lvert h\rvert)$ (right of 1)	1.69%	3.61%	6.03%
$E(-\lvert h\rvert)$ (left of 1)	1.96%	4.50%	8.12%

Solving $E = 5\%$ numerically: right side at $h \approx +0.36$, left side at $h \approx -0.32$. So the relative error **first exceeds** 5% **at** $|h| \approx 0.32$, on the left. The linear model is within 5% on roughly [0.68, 1.36]. *Testing only $h > 0$ gives 0.36, which is the classic mistake. The left side breaks first because the curvature isn't symmetric. Always check both sides.*

SOLUTION // Exercise 14a: one gradient step decreases f

Move: plug the step into the first-order Taylor model.

1. Taylor with $h = -\eta f'(x)$: $f(x+h) = f(x) + f'(x)h + o(|h|)$.
2. So $f(x - \eta f'(x)) = f(x) - \eta(f'(x))^2 + o(\eta)$.
3. If $f'(x) \neq 0$, the main change $-\eta(f'(x))^2$ is **strictly negative**, and it's proportional to η, while the error is $o(\eta)$ (negligible compared to η). So for η small enough, the error can't cancel the decrease: $f(x - \eta f'(x)) < f(x)$.

SOLUTION // Exercise 14b: a safe step size

Fix the hypothesis first. To control the Taylor error we need a bound on the *curvature*: assume $|f''| \leq L$ on the interval (equivalently, f' is L-Lipschitz). A bound like "$|f'(x)| \leq L$" is not enough: a bound on f' alone says nothing about how fast the slope changes.

1. Taylor with remainder: for some ξ between x and $x - \eta f'(x)$,
$$f(x - \eta f'(x)) = f(x) - \eta(f'(x))^2 + \tfrac{1}{2} f''(\xi)\, \eta^2 (f'(x))^2.$$
2. Error: $|E| \leq \frac{1}{2} L \eta^2 (f'(x))^2$. Predicted decrease: $D = \eta(f'(x))^2$.
3. Ratio: $\dfrac{|E|}{D} \leq \dfrac{L\eta}{2}$.
4. Want the error to be at most a fraction $c < 1$ of the decrease: $\eta \leq \dfrac{2c}{L}$. With $c = \frac{1}{2}$: $\boxed{\eta \leq \dfrac{1}{L}}$.
5. Bonus: for any $\eta < \frac{2}{L}$, $f(x - \eta f'(x)) \leq f(x) - \eta(f'(x))^2(1 - \frac{L\eta}{2}) < f(x)$: still a guaranteed decrease.

Example: $f(x) = 3x^2$ has $f'' = 6 = L$. With $\eta = \frac{1}{6}$ from $x = 1$: $x - \eta f'(x) = 1 - \frac{1}{6} \cdot 6 = 0$, the minimum in one step. With $\eta = \frac{1}{3} = \frac{2}{L}$: you land at -1, f unchanged (the edge). With $\eta > \frac{1}{3}$: f goes **up**.

SOLUTION // Exercise 14c: gradient of $f(w) = \|Xw - y\|_2^2$

1. Expand: $f(w) = w^\top X^\top X w - 2w^\top X^\top y + y^\top y$.
2. Rules: $\nabla(w^\top A w) = 2Aw$ for symmetric A; $\nabla(w^\top b) = b$; constants vanish.
3. $\nabla f(w) = 2X^\top X w - 2X^\top y = \boxed{2X^\top(Xw - y)}$ (same as the chain-rule derivation in Section 10).

4. Component i: $\frac{\partial f}{\partial w_i} = 2\langle x_i, Xw - y\rangle$, with x_i the i-th **column** of X (feature i over all samples). It's large when feature i is strongly correlated with the current residual: nudging w_i would change the loss a lot. It's 0 when the residual is orthogonal to that feature (Lecture 5!).

Numbers: $X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$, $y = (1, 2, 2)$, $w = (0, 0)$: residual $Xw - y = (-1, -2, -2)$, so $\nabla f = 2\,(-1 - 2 - 2,\ 0 - 2 - 4) = (-10, -12)$. Both weights want to increase, w_2 a bit more.

11.4 Bonus: extra practice questions

SOLUTION // $a_n, b_n = O(1/n)$: is $a_n + b_n = O(1/n)$? Is $a_n b_n = O(1/n)$?

$|a_n| \le \frac{C_1}{n}$ for $n \ge N_1$ and $|b_n| \le \frac{C_2}{n}$ for $n \ge N_2$. For $n \ge \max(N_1, N_2)$:

1. $|a_n + b_n| \le \frac{C_1 + C_2}{n}$: yes, $O(1/n)$.
2. $|a_n b_n| \le \frac{C_1 C_2}{n^2} \le \frac{C_1 C_2}{n}$: yes, and even better, $O(1/n^2)$.

SOLUTION // Jacobian of $F(x, y) = (xe^y,\ \sin(x + y))$ at $(0, 0)$

$J_F = \begin{pmatrix} e^y & xe^y \\ \cos(x + y) & \cos(x + y) \end{pmatrix}$, so $J_F(0, 0) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. With $F(0, 0) = (0, 0)$: $F(h_1, h_2) \approx (h_1,\ h_1 + h_2)$. Check at $h = (0.1, 0.1)$: true $(0.1e^{0.1}, \sin 0.2) \approx (0.1105, 0.1987)$ vs approx $(0.1, 0.2)$ ✓.

SOLUTION // Best rank-1 approximation and its Frobenius error

$A_1 = \sigma_1 u_1 v_1^\top$, and $\|A - A_1\|_F = \sqrt{\sigma_2^2 + \cdots + \sigma_r^2}$ (and $\|A - A_1\|_2 = \sigma_2$).

SOLUTION // $|x|$ and $x|x|$ at 0

1. $|x|$: continuous ($||x| - 0| = |x| \to 0$). Not differentiable: $\frac{|h|}{h} = \pm 1$ depending on the side.
2. $x|x|$: continuous. Differentiable: $\frac{h|h| - 0}{h} = |h| \to 0$, so the derivative at 0 is 0. Its derivative everywhere is $2|x|$, which is continuous, but not differentiable at 0. So $x|x|$ is C^1 but not C^2.

12 Final boss: the cheat sheet

12.1 "If they ask you this, do that"

If asked...	**Do this**
$\|A\|_1$ / $\|A\|_\infty$	Max absolute **column** sum / max absolute **row** sum. Give the vector that attains it (e_j / sign vector).
$\|A\|_2$	$\sqrt{\lambda_{\max}(A^\top A)} = \sigma_1$. Symmetric or normal: $\max \|\lambda_i\|$. Attained at v_1.
Prove a bound on $\|AB\|$ or $\|Ax\|$	Submultiplicativity $\|AB\| \le \|A\|\|B\|$; subordinate property $\|Ax\| \le \|A\|\|x\|$.
Best rank-k approximation	Truncate the SVD. Errors: σ_{k+1} (spectral), $\sqrt{\sum_{i>k} \sigma_i^2}$ (Frobenius).
Least squares, any matrix	$x^\star = A^+ b$, $A^+ = V\Sigma^+ U^\top$; minimal-norm solution; $Ax^\star$ = projection of b.
Prove $x_n \to L$	Bound $\|x_n - L\|$ by something simple, solve for $N(\varepsilon)$, write "Let $\varepsilon > 0$, take $N = \dots$".
Prove divergence	Two subsequences with different limits, or unbounded.
Prove a sequence converges without the limit	Monotone + bounded, or Cauchy.
Compare growth	Limit of the ratio: $0 \Rightarrow o$, constant $\Rightarrow \Theta$. Ladder $\log n \ll n^\alpha \ll \beta^n$.
Sum like $\sum f(k)$	Compare with $\int f$ (decreasing f): $H_n = \Theta(\log n)$; $\sum k^{-p} < \infty \iff p > 1$.
ε–η limit proof	Write $\|f(x) - \ell\| \le C\|x - a\|$, take $\eta = \varepsilon / C$.
Compute a limit	Plug in; $\frac{0}{0}$: factor/squeeze/known limit; $\frac{c}{0}$: blow-up; piecewise: both sides.
Is f continuous at a?	Left limit = right limit = $f(a)$.
Linearise f	$\ell_1(x) = f(x_0) + f'(x_0)(x - x_0)$; error $\approx \frac{1}{2}\|f''\|h^2$; test **both** sides.
Gradient / directional derivative	Partial derivatives (others constant); **normalise** u; $D_u f = \langle \nabla f, u \rangle \le \|\nabla f\|$.
Jacobian	Row i = gradient of F_i; shape outputs $\times$ inputs; $F(x + h) \approx F(x) + J_F h$.
Chain rule	$J_{F \circ G}(x) = J_F(G(x))\, J_G(x)$: check shapes, evaluate J_F at $G(x)$.
Gradient of $\|Xw - y\|^2$	$2X^\top(Xw - y)$.
Safe GD step size	If $\|f''\| \le L$: $\eta \le 1/L$ (error $\le$ half the decrease); $\eta < 2/L$ still decreases.

12.2 Formula sheet

Operator norm	$\|A\|_{\text{op}} = \sup_{\|x\|=1}\|Ax\| = \sup_{\|x\|\le 1}\|Ax\|$; $\|Ax\| \le \|A\|_{\text{op}}\|x\|$; $\|AB\|_{\text{op}} \le \|A\|_{\text{op}}\|B\|_{\text{op}}$
$p = 1, \infty$	$\|A\|_1 = \max_j \sum_i \|a_{ij}\|$, $\|A\|_\infty = \max_i \sum_j \|a_{ij}\|$
Spectral	$\|A\|_2 = \sigma_1 = \sqrt{\lambda_{\max}(A^\top A)}$; $\|A\|_2 \le \|A\|_F \le \sqrt{r}\,\|A\|_2$; $\|A\|_F^2 = \sum \sigma_i^2$
Eckart–Young	$\|A - A_k\|_2 = \sigma_{k+1}$, $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$
Pseudo-inverse	$A^+ = V\Sigma^+ U^\top$, $\kappa_2 = \sigma_1/\sigma_r$
Convergence	$\forall \varepsilon\, \exists N : n \ge N \Rightarrow \|x_n - x\| < \varepsilon$; Cauchy $\iff$ convergent in $\mathbb{R}^d$
Asymptotics	O: $\|x_n\| \le C\|y_n\|$; o: $x_n/y_n \to 0$; Θ: $c_1\|y_n\| \le \|x_n\| \le c_2\|y_n\|$
Harmonic, p-series	$\log(n+1) \le H_n \le 1 + \log n$; $\sum k^{-p} < \infty \iff p > 1$
Limit	$\forall \varepsilon\, \exists \eta : 0 < \|x - a\| < \eta \Rightarrow \|f(x) - \ell\| < \varepsilon$; $\frac{\sin x}{x} \to 1$
Taylor (1st order)	$f(x+h) = f(x) + f'(x)h + o(\|h\|)$; $f(x+h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$
Directional derivative	$D_u f = \langle \nabla f, u\rangle$, $\|u\| = 1$; max $= \|\nabla f\|$ along $\nabla f/\|\nabla f\|$
Jacobian	$[J_F]_{ij} = \partial F_i/\partial x_j$, size $m \times n$; $F(x+h) = F(x) + J_F(x)h + o(\|h\|)$
Chain rule	$J_{F\circ G}(x) = J_F(G(x))J_G(x)$; $\nabla(f \circ G) = J_G^\top \nabla f(G)$
GD	$f(x - \eta f') \approx f(x) - \eta(f')^2$; $\eta \le 1/L$ if $\|f''\| \le L$

12.3 Hall of Ls

COMMON L (don't do this) // Do not

1. Mix up $\|A\|_1$ (columns) and $\|A\|_\infty$ (rows).
2. Say $\|A\|_2 = \max|\lambda_i|$ for a non-normal matrix. Counterexample: $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, eigenvalues 0, norm 1.
3. Confuse $\|A\|_2$ (one singular value) with $\|A\|_F$ (all of them).
4. Think $\|AB\| = \|A\|\|B\|$. It's only $\le$.
5. Run PCA/SVD on uncentred data: the first direction just finds the mean.
6. Forget to flip the inequality when dividing by $\log \rho < 0$.
7. Say "bounded, so convergent" ($(-1)^n$), or "oscillates, so divergent" ($(-1)^n/n$).
8. Treat $\frac{0}{0}$ as a value. And treat $\frac{c}{0}$ ($c \ne 0$) as indeterminate: it's a blow-up.
9. Declare continuity from equal one-sided limits without checking $f(a)$.
10. Think continuous $\Rightarrow$ differentiable ($|x|$).
11. Forget to normalise u in a directional derivative.
12. Write a Jacobian as inputs $\times$ outputs. It's outputs $\times$ inputs.
13. Evaluate the outer Jacobian at x instead of at $G(x)$.

14. Check a linear approximation on one side only (Exercise 13d).

12.4 The whole lecture in 10 lines

TL;DR // Screenshot this

1. $\|A\|_{\text{op}}$ = max stretch of a unit vector = smallest Lipschitz constant of $x \mapsto Ax$; submultiplicative.
2. $\|A\|_1$ = max column sum, $\|A\|_\infty$ = max row sum, $\|A\|_2 = \sigma_1$.
3. SVD = sum of rank-one pieces $\sigma_i u_i v_i^\top$; keep the top k = best rank-k (Eckart–Young), errors σ_{k+1} and $\sqrt{\sum_{i>k} \sigma_i^2}$.
4. Pseudo-inverse solves least squares for any matrix, with the minimal-norm answer; $\kappa = \sigma_1/\sigma_r$ measures sensitivity.
5. Convergence: ε–N. In $\mathbb{R}^d$: Cauchy $\iff$ convergent; any norm, same verdict; monotone + bounded converges.
6. $O \leq$, $o \ll$, $\Theta \approx$. $\log n \ll n^\alpha \ll \beta^n$; $H_n = \Theta(\log n)$.
7. Limits: ε–η; removable / jump / infinite; continuity = limit equals value.
8. $f(x+h) \approx f(x) + f'(x)h$: local only; a gradient step decreases f if η is small ($\eta \leq 1/L$).
9. ∇f points steepest uphill, $\perp$ level sets; $D_u f = \langle \nabla f, u \rangle$.
10. Jacobian = best linear zoom ($m \times n$); chain rule multiplies Jacobians = backprop; products of norms explain exploding/vanishing gradients.

Matrices and calculus, finally in the same room. Go get that exam, bestie.